labs(), Annotations & Storytelling Layers

Workshop 2 — Module 8

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Data Visualization for Data Scientists
UNYT Tirana

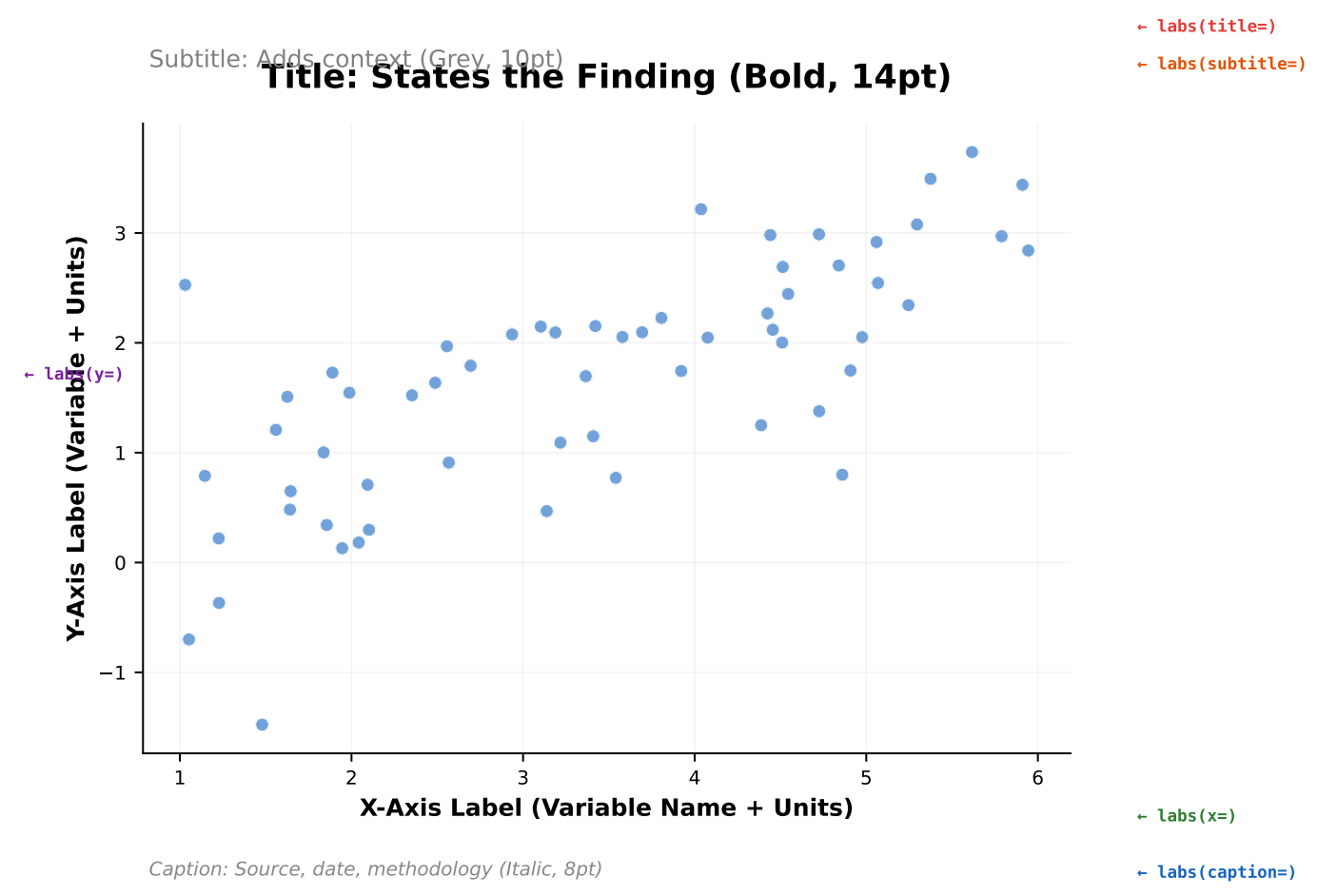
Module Roadmap

1. `labs()`: title, subtitle, caption, axis labels
2. Title patterns: descriptive, declarative, question
3. Annotation types: text, arrow, reference line, band, highlight
4. `annotate()` vs `geom_text()` vs `geom_text_repel()`
5. Storytelling layers: progressive reveal
6. Complete annotated chart in R and Python

Learning Outcome
You will transform a data display into a data story by adding labels, annotations, and reference marks that guide the reader to the intended finding.

labs(): The Label Layer

Anatomy of labs()



labs() in R vs Python

R (ggplot2)

```r
ggplot(df, aes(x, y, color = g))
    +
  geom_point() +
  labs(
    title = "Q4 Revenue Surged
        35%",
    subtitle = "Quarterly revenue
        ,
                East region, 2024
                    ",
    caption = "Source: Internal
        DB",
    x = "Displacement (L)",
    y = "Highway MPG",
    color = "Vehicle Class"
  )
```

Python (matplotlib)

```python
ax.set_title("Q4 Revenue Surged
    35%",
    fontsize=14, fontweight="bold
        ")
fig.text(0.13, 0.91,
    "Quarterly revenue, East,
        2024",
    fontsize=10, color="grey")
ax.set_xlabel("Displacement (L)")
ax.set_ylabel("Highway MPG")
fig.text(0.13, 0.02,
    "Source: Internal DB",
    fontsize=8, fontstyle="italic
        ",
    color="#888")
ax.legend(title="Vehicle Class")
```

Pro-Tip

In ggplot2, `labs()` sets all labels in one call. In matplotlib, title/xlabel/ylabel are on the axes, but subtitle and caption require `fig.text()` with manual coordinates.

Title Patterns

Three Title Strategies

Three Title Patterns: Descriptive → Declarative → Question

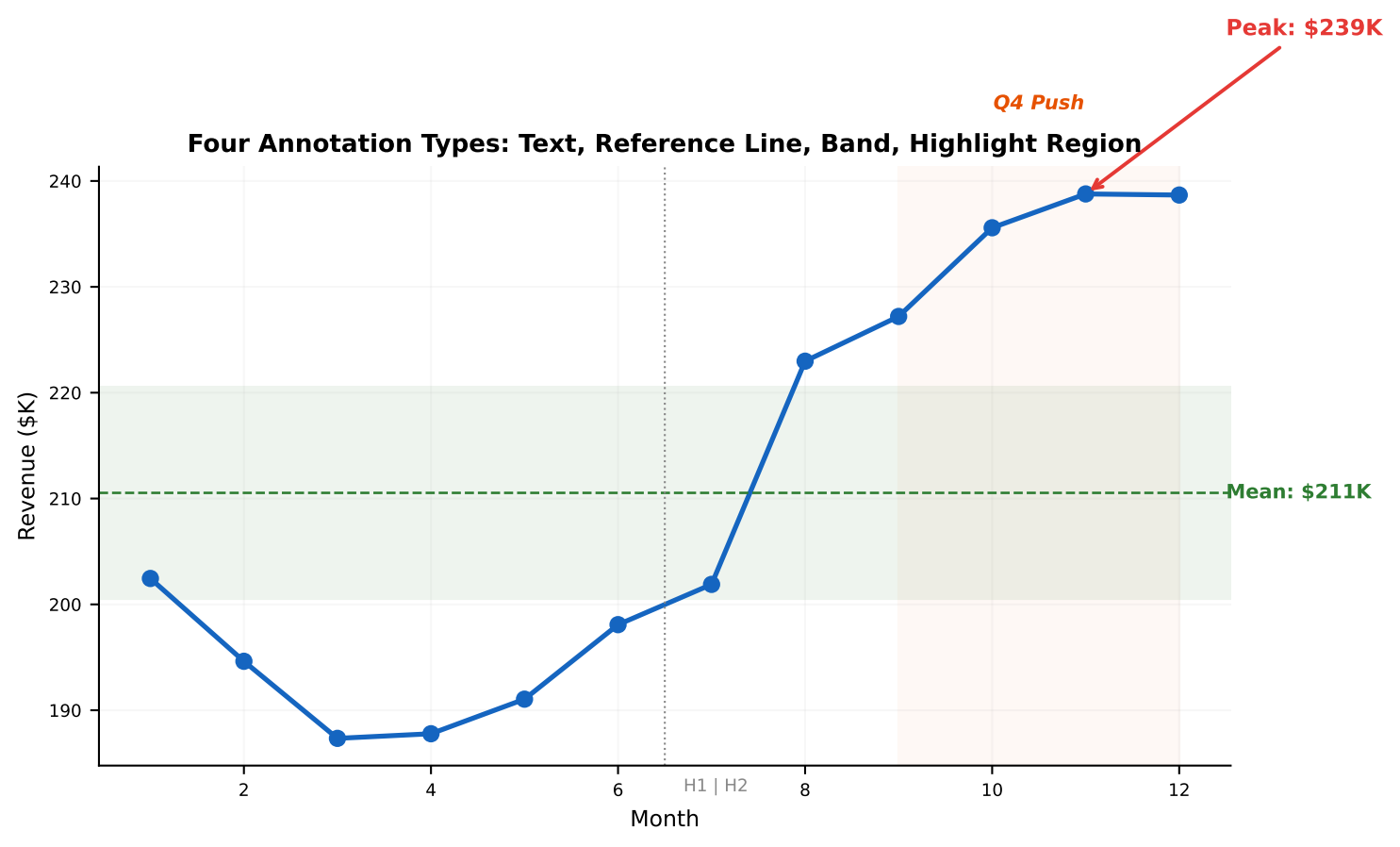
Pattern	Example	When to Use	Strength
Descriptive (what it shows)	**Revenue by Quarter**	Neutral, exploratory Use when audience draws own conclusions	
Declarative (what it means)	**Q4 Revenue Surged 35% to Record $395K**	Persuasive, directed Use when you have a clear message	
Question (what to explore)	**Which Quarter Drove the Revenue Spike?**	Engaging, Socratic Use for interactive or exploratory contexts	

Data Insight

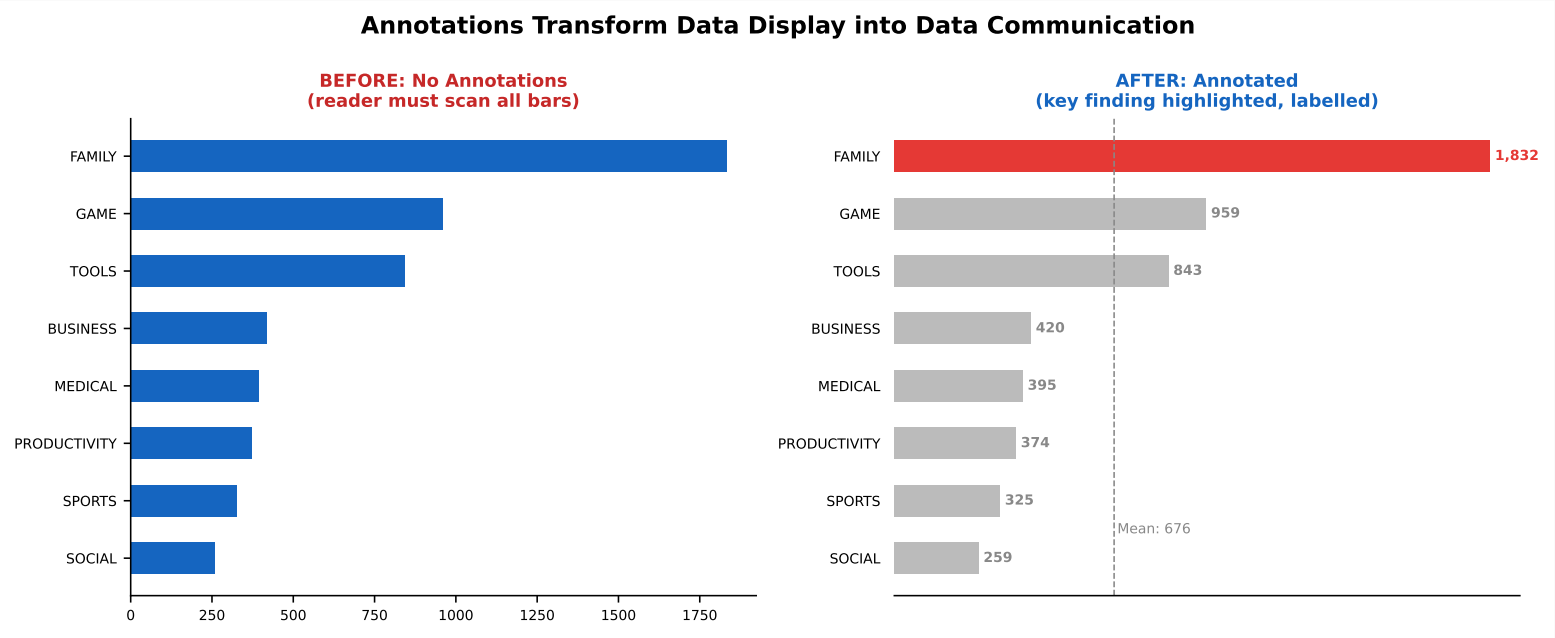
For executive communication, use **declarative** titles that state the finding as a sentence. For exploratory analysis, use **descriptive** titles. For teaching or interactive contexts, use **question** titles that invite investigation.

Annotation Types

Four Annotation Types in One Chart



Before & After: Annotations Add Meaning



Annotations in R (ggplot2)

```r
ggplot(df, aes(x = month, y = revenue)) +
  geom_line(color = "#1565C0", linewidth = 1.2) +

  # 1. Text callout with arrow
  annotate("text", x = 8, y = 250, label = "Peak: $250K",
           fontface = "bold", color = "#E53935", size = 4) +
  annotate("segment", x = 8, y = 245, xend = 7, yend = 235,
           arrow = arrow(length = unit(0.2, "cm")), color = "#E53935
               ") +

  # 2. Reference line (mean)
  geom_hline(yintercept = mean(df$revenue),
             linetype = "dashed", color = "grey50") +

  # 3. Reference band (target range)
  annotate("rect", xmin = -Inf, xmax = Inf, ymin = 200, ymax = 220,
           alpha = 0.08, fill = "#2E7D32") +

  # 4. Event marker (vertical line)
  geom_vline(xintercept = 6.5, linetype = "dotted", color = "#E65100
      ") +
  annotate("text", x = 7, y = min(df$revenue), label = "Policy
      change",
           size = 3, color = "#E65100")
```

Annotations in Python (matplotlib)

```
ax.plot(months, revenue, "-o", color="#1565C0", lw=2)

# 1. Text callout with arrow
ax.annotate("Peak: $250K",
    xy=(7, 235), xytext=(8, 250),            # data coords
    fontsize=9, fontweight="bold", color="#E53935",
    arrowprops=dict(arrowstyle="->", color="#E53935", lw=1.5))

# 2. Reference line (mean)
ax.axhline(y=np.mean(revenue), color="grey", lw=1, linestyle="--")

# 3. Reference band (target range)
ax.axhspan(200, 220, alpha=0.08, color="#2E7D32")

# 4. Event marker (vertical line)
ax.axvline(x=6.5, color="#E65100", lw=0.8, linestyle=":")
ax.text(6.7, min(revenue), "Policy change", fontsize=7, color="#
    E65100")
```

annotate() vs geom_text()

Three Labelling Functions

annotate() vs geom_text() vs geom_text_repel()

Function	Positioning	Best For	Note
`annotate()`	Fixed position (data or npc coords)	Single callout, reference label, arrow + text	One annotation per call
`geom_text() / geom_label()`	Data-driven position (one label per row)	Direct labelling all data points	One label per observation
`ggrepel:: geom_text_repel()`	Data-driven + auto-repulsion	Direct labelling without overlap	Anti-collision algorithm

geom_text_repel: Anti-Overlap Labels

R

```r
library(ggrepel)

ggplot(top10, aes(x = reviews,
    y = rating, label = app)) +
  geom_point(alpha = 0.6) +
  # Labels repel from each other
  geom_text_repel(
    size = 3,
    max.overlaps = 15,
    segment.color = "grey70",
    segment.size = 0.3) +
  scale_x_log10() +
  theme_minimal()
```

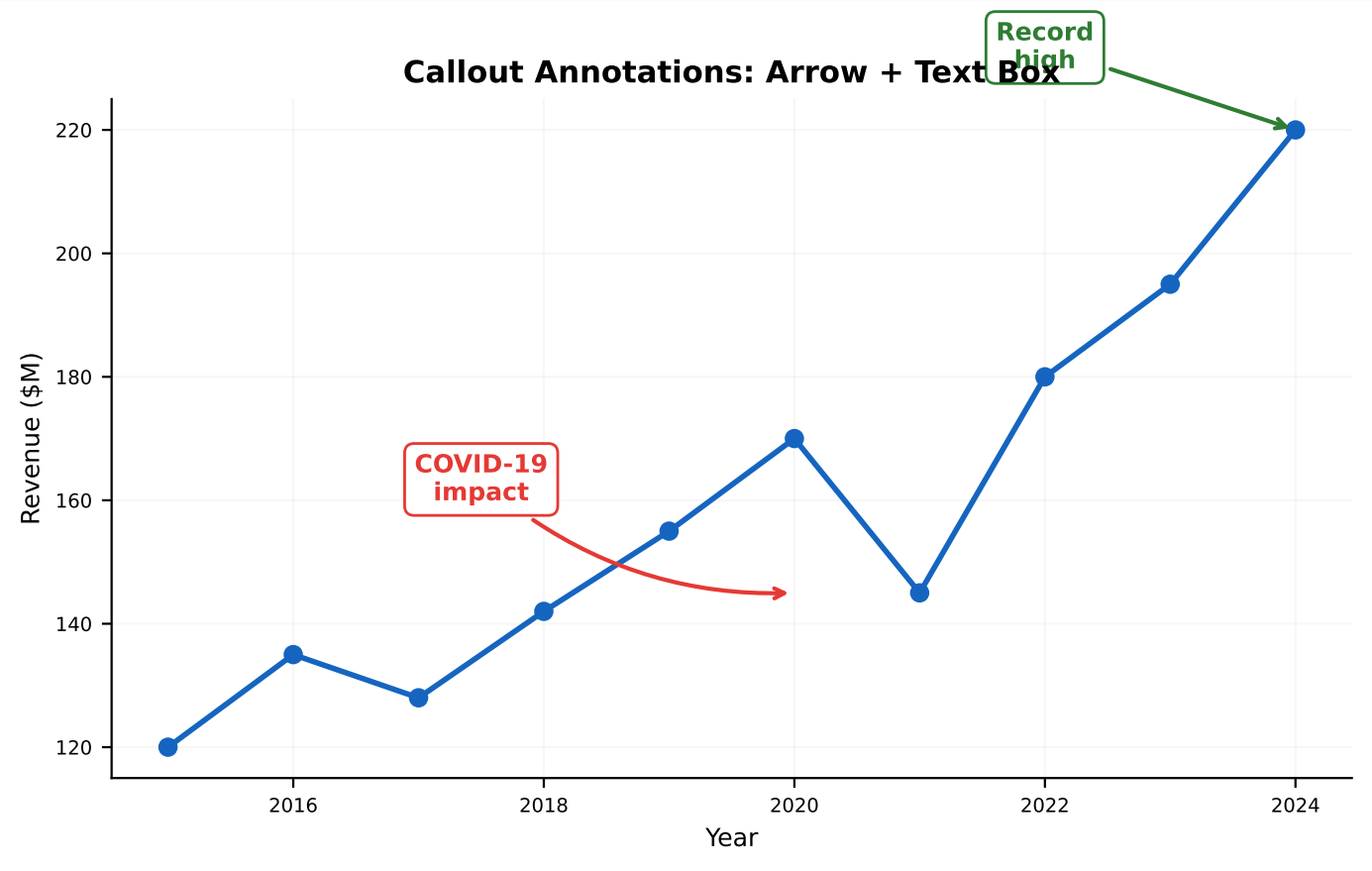
Python

```python
from adjustText import
     adjust_text

ax.scatter(reviews, rating, s=20)
texts = [ax.text(r, rat, name,
    fontsize=7)
    for r, rat, name in
    zip(reviews, rating, names)]

# Repel overlapping labels
adjust_text(texts,
    arrowprops=dict(
        arrowstyle="-",
        color="grey",
        lw=0.5))
```

Install: `pip install adjustText`

Callout Annotations

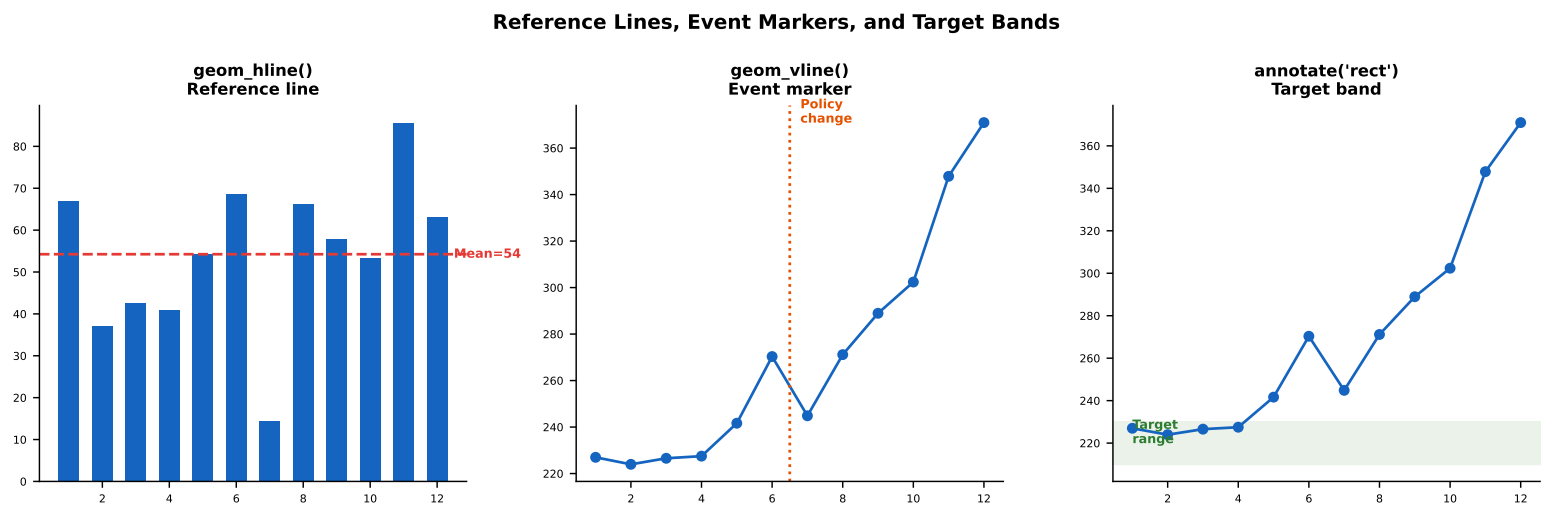
Arrow + Text Box: The Callout Pattern



Pro-Tip

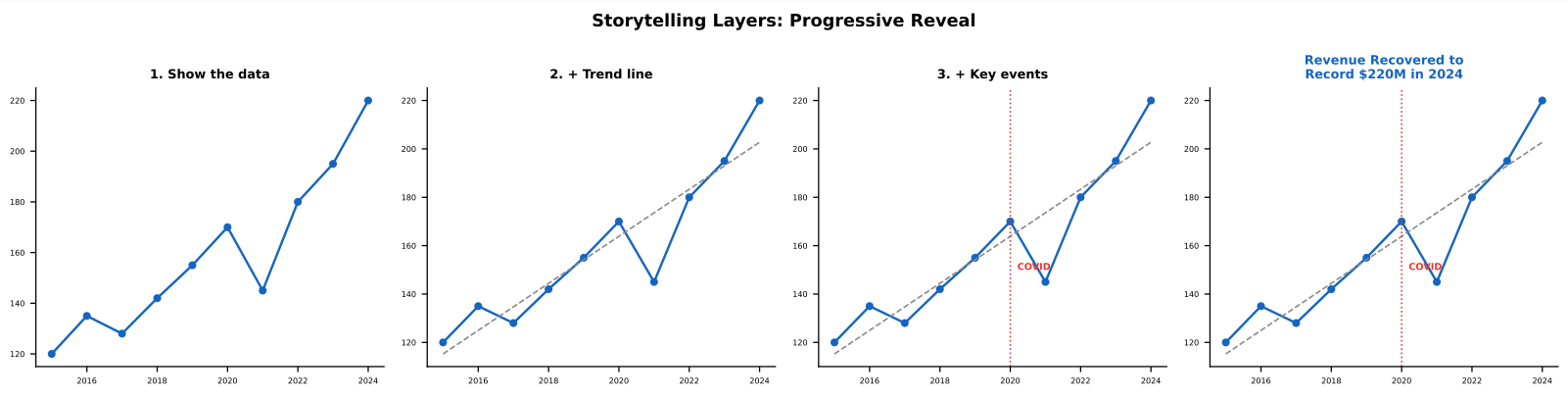
Use callouts sparingly — one or two per chart maximum. Each callout should highlight a **specific finding**, not just label a data point. "COVID-19 impact" is a finding; "2020" is just a label.

Reference Lines, Event Markers, Target Bands



Storytelling Layers

Progressive Reveal: Four Layers of Meaning



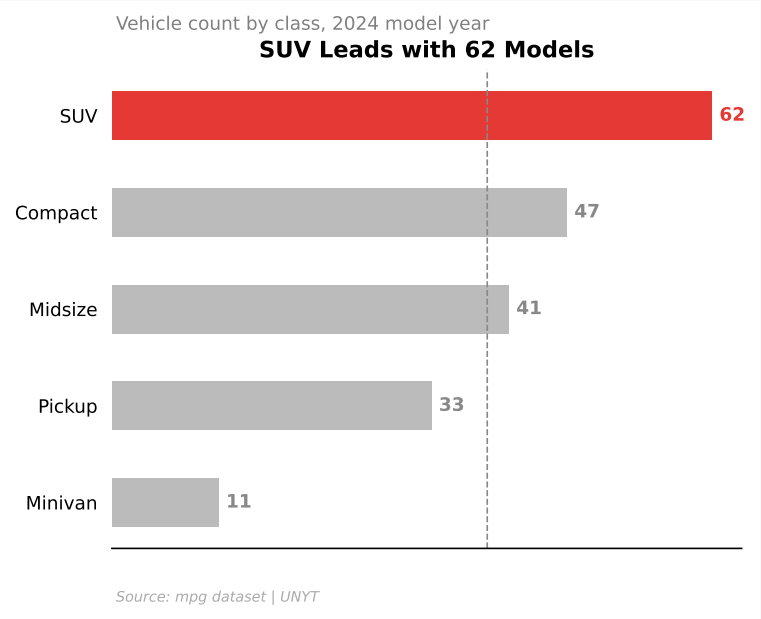
Data Insight

Each layer adds one element of the story: (1) raw data, (2) trend context, (3) key events, (4) the finding as a title. This layering technique works equally well for presentations (reveal one slide at a time) and static reports (the final chart contains all layers).

Complete Examples

R: Fully Annotated Bar Chart

```r
df |>
  mutate(highlight =
    cat == "SUV") |>
  ggplot(aes(
    x = fct_reorder(cat, n),
    y = n,
    fill = highlight)) +
  geom_col(width = 0.5,
    show.legend = FALSE) +
  geom_text(aes(label = n),
    hjust = -0.2, size = 3.5,
    fontface = "bold") +
  scale_fill_manual(
    values = c("TRUE" = "#E53935",
               "FALSE" = "#BBBBBB")
                    ) +
  geom_hline(yintercept = mean_n,
    linetype = "dashed",
    color = "grey50") +
  coord_flip() +
  labs(
    title = "SUV Leads with 62",
    subtitle = "Vehicle count, 2024
        ",
    caption = "Source: mpg | UNYT",
    x = NULL, y = NULL) +
```

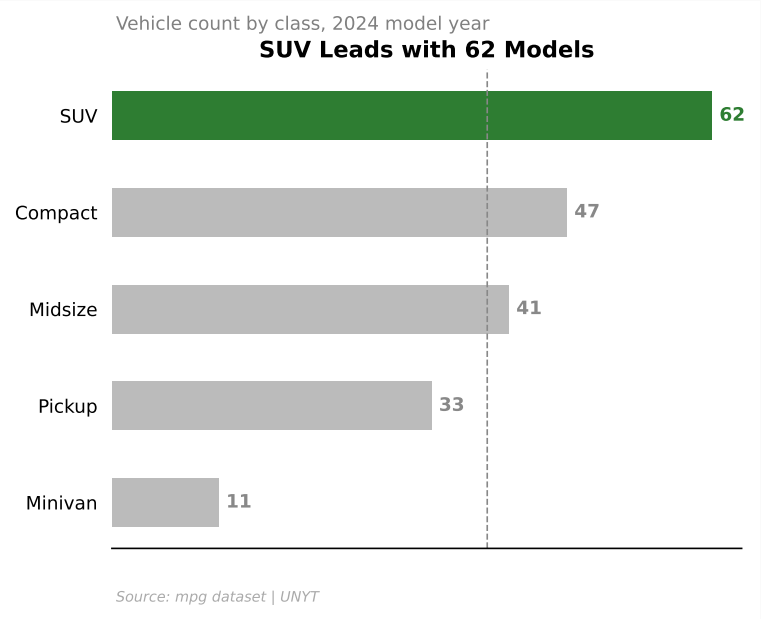


Annotation checklist:

1. Declarative title
2. Contextual subtitle
3. Source caption
4. Direct bar labels
5. Grey-accent colour
6. Mean reference line

Python: Same Annotated Chart

```python
fig, ax = plt.subplots(figsize
    =(6,5))

colors = ["#BBBBBB"] * 5
colors[-1] = "#2E7D32"  # accent

ax.barh(sorted_cats, sorted_vals,
    color=colors, height=0.5)

# Direct labels
for i, v in enumerate(sorted_vals):
    ax.text(v + 0.8, i, str(v),
        va="center", fontsize=9,
        fontweight="bold",
        color="#2E7D32"
            if i == 4 else "#888")

# Reference line
ax.axvline(x=np.mean(sorted_vals),
    color="#888", lw=0.8, ls="--")

# Title + subtitle + caption
ax.set_title("SUV Leads with 62",
    fontsize=11, fontweight="bold")
fig.text(0.13, 0.88,
    "Vehicle count by class, 2024 model year
```



Synthesis

Annotation Checklist

Before finalising any chart, verify:

1. **Title**: Does it state the finding (declarative) or describe the chart (descriptive)?
2. **Subtitle**: Does it add context (time period, filter, data source)?
3. **Caption**: Does it credit the source and note methodology?
4. **Axis labels**: Do they include units ($, %, kg)?
5. **Direct labels**: Are key values shown on the chart (eliminating need to read axes)?
6. **Reference marks**: Is there a mean line, target band, or event marker for context?
7. **Accent colour**: Is the key finding visually highlighted?

Key Takeaways

1. **labs()** sets title, subtitle, caption, and axis labels. Use declarative titles for executive audiences.
2. Four annotation types: **text callout** (arrow + label), **reference line** (mean, target), **band** (range), **highlight region** (event period).
3. **annotate()** for fixed-position single annotations; **geom_text()** for data-driven labels; **geom_text_repel()** for anti-overlap.
4. **Storytelling layers** build meaning progressively: data $\rightarrow$ trend $\rightarrow$ events $\rightarrow$ finding.
5. Every production chart needs: title, subtitle, caption, units, and at least one annotation that guides the reader.

Essential Readings

- Knaflic, C. N. (2015). *Storytelling with Data*, Chapter 4 (Focus Your Audience's Attention).
- Schwabish, J. (2021). *Better Data Visualizations*, Chapter 5 (Annotations).
- Wickham, H. (2016). *ggplot2*, Chapter 8.3 (Annotations).
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 22 (Titles, Captions, and Tables).

Next Module: Tidy Data & Data Wrangling for Viz (W02-M09)